

## Analysis First-Year Exam, May 2024, Part 1

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Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Assume without proof that if  $(a_n)$  and  $(b_n)$  are bounded real sequences then

$$\limsup(a_n + b_n) \leq \limsup a_n + \limsup b_n.$$

(i) Show by example that this inequality can be strict.

(ii) Prove that equality holds if  $(a_n)$  is convergent.

2. Let  $A$  be a subset of the metric space  $M$ . Assume that  $A$  satisfies the following condition:

$$\forall X \subseteq M \left( A \cap X = \emptyset \Rightarrow A \cap \overline{X} = \emptyset \right).$$

Does it follow that  $A$  is open? Proof or counterexample.

3. Let  $A$  and  $B$  be nonempty compact subsets of the metric space  $M$ . Prove that there exist  $a \in A$  and  $b \in B$  such that

$$d(a, b) = \inf\{d(x, y) : x \in A, y \in B\}.$$

4. Does there exist a one-to-one continuous function from the unit square  $\square = [0, 1] \times [0, 1]$  to  $\mathbb{R}$ ? If so, exhibit one; if no, prove not.
5. Let  $f : (0, \infty) \rightarrow \mathbb{R}$  be differentiable and let  $b \in \mathbb{R}$ . Prove that if  $f'(t) \rightarrow b$  as  $t \rightarrow \infty$  then  $f(t)/t \rightarrow b$  as  $t \rightarrow \infty$ . [It might help to choose and fix  $a > 0$ .]